



Semester Project On the Modelling of Modular Multilevel Converters

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1 Introduction

Nearly 40% of global greenhouse gas emissions stem from energy-intensive sectors, including the production of electricity and heat [1]. A major share of these emissions is caused by the continued reliance on fossil-fuel combustion for generating electricity. In 2016, the Paris Agreements committed 195 countries to limiting the rise in global average temperature to well below 2 °C [2]. Achieving this target requires a substantial transformation of existing power systems and has accelerated the shift toward net-zero energy infrastructures. Such systems depend largely on Inverter Based Resources (IBRs), like solar and wind, rather than the traditional synchronous machines powered by fossil fuels [3].

One of the technologies enabling the conversion to these net-zero grids is High Voltage Direct Current (HVDC), which uses direct current instead of conventional High Voltage Alternating Current (HVAC) to transport power. It brings along a multitude of benefits compared to AC transmission [4], with the main applications being [5]:

- long-distance power transmission lines;
- submarine cable transmission (connections with lengths greater than 50 km, usually used for offshore wind power plants);
- interconnection of systems that operate asynchronously;
- multi-terminal HVDC systems.

For synthesizing the voltages required in HVDC systems, standard two- or three-level Voltage Source Converters (VSCs) often face fundamental restrictions in their topology. A two-level VSC uses one DC voltage source to create an averaged AC output voltage through PWM. This requires switching the full DC-link voltage through a single semiconductor device, which leads to relatively high switching losses and inherently limits the maximum output voltage to the voltage rating of that device [6].

Modular Multilevel Converters (MMCs) were developed to overcome these limitations. MMCs generate a staircase-shaped output waveform using multiple independent voltage sources formed by capacitor banks in the sub-modules. This enables the converter to synthesize high voltages in a scalable and modular way. As a result, MMCs provide several key advantages [6]:

- lower harmonic distortion, reducing the need for bulky AC filters;
- significantly lower switching losses;
- reduced voltage stress on individual switching devices;
- high modularity and fault tolerance, improving system reliability.

Despite these advantages, MMCs also introduce behaviours that do not appear in conventional VSCs. Because each arm contains many sub-module capacitors, the converter stores energy in a distributed way rather than in a single DC-link capacitor.

This leads to internal power flows that exist independently of the external AC–DC power exchange.

As a result, phenomena such as circulating currents, second-harmonic energy oscillations, and energy imbalances between the upper and lower arms naturally arise. These effects influence the capacitor voltages, dynamic stability, and the quality of the synthesized AC waveform. If left uncontrolled, they can lead to increased losses, voltage drift, and reduced converter performance.

For applications such as STATCOMs, where the converter must react quickly to grid conditions and provide accurate reactive power support, maintaining stable internal energy dynamics is therefore essential. The objective of this work is to construct a fully functional three-phase average-model MMC in MATLAB/Simulink, developing the model step by step and gaining a detailed understanding of the purpose and behavior of each element introduced along the way.

The remainder of this document is organised as follows. Chapter 2 introduces the background and motivation. Chapter 3 explains the modelling and control theory used throughout the work. Chapter 4 describes how these elements are implemented in Simulink. The resulting behaviour is shown in Chapter 5, and final remarks and suggestions for future work are given in Chapter 6.

2 Background

2.1 In Comparison with Two-Level VSC

Classical two-level and three-level Voltage Source Converters for HVDC generate the AC voltage by switching the full DC-link across the phase leg. In a two-level design, the phase voltage can only be $+V$ or $-V$. In a three-level design, the possible values are $+V$, 0 , or $-V$. Pulse Width Modulation (PWM) is used to modulate these voltage levels into an average sinusoidal waveform. The difference between a perfect sinusoidal waveform and the synthesized one appears as switching ripple and harmonics, which need to be filtered.

Because IGBTs have limited blocking capability, many devices must be placed in series to reach HVDC voltage levels. When this is done, all series-connected devices must switch at exactly the same moment to ensure safe voltage sharing. This absolute simultaneity requirement complicates the gate-drive design and protection. It also becomes more difficult at higher voltages, with more devices stacked. Combined with the large voltage steps at the output, this leads to high dv/dt , higher switching losses, and large filters [7].

These limitations led to the development of multilevel converters, where the AC voltage is built from many small steps instead of one large transition. Figure 2 shows that as the number of voltage levels increases, the output waveform becomes smoother and needs less filtering.

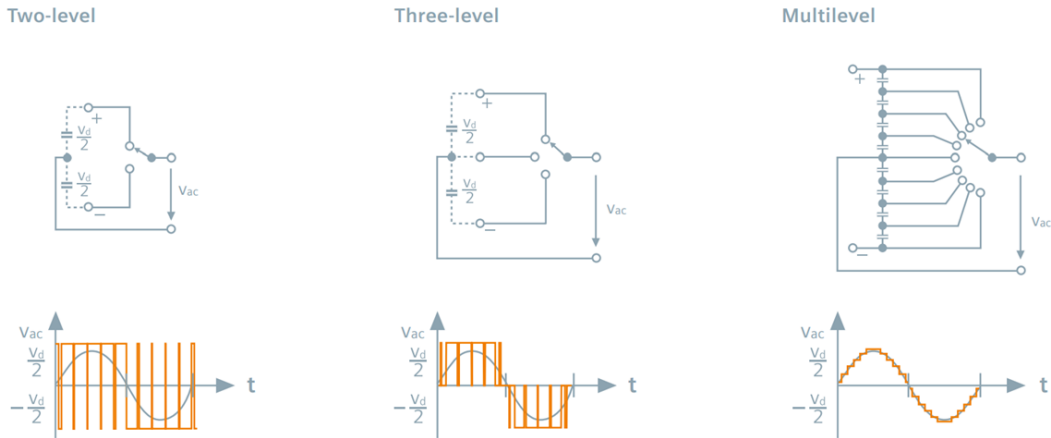


Figure 2: Two, Three, and Multi-level Converter Diagram

2.2 MMC Overview

The Modular Multilevel Converter was developed to overcome the limitations of two-level and three-level VSCs. Instead of switching a single large voltage step, an MMC builds the AC voltage from many small steps using sub-modules connected in series within each arm. Each sub-module contains a capacitor and can either be inserted or bypassed, allowing the arm voltage to increase or decrease in small increments. As a result, the output waveform becomes much smoother, and the required AC filtering is significantly reduced.

A distinguishing feature of the MMC is its distributed energy storage: each sub-module capacitor stores part of the total energy, rather than relying on a single DC-link capacitor. This introduces internal power flows and circulating currents that must be controlled but also gives the converter additional flexibility. Another important advantage is scalability: since each device only blocks a fraction of the total voltage, the converter can reach high power and high voltage levels simply by adding more sub-modules.

Figure 3 shows an overview of the MMC topology. The upper and lower arms each contain a series connection of sub-modules and an arm inductor. By inserting different numbers of sub-modules in each arm, the converter produces two staircase voltages that combine into the AC phase voltage. On the right side of the figure, the typical shape of the upper arm voltage, lower arm voltage, and resulting AC output voltage is illustrated [8], [9]. The output approaches a sinusoid without relying on high switching frequency.

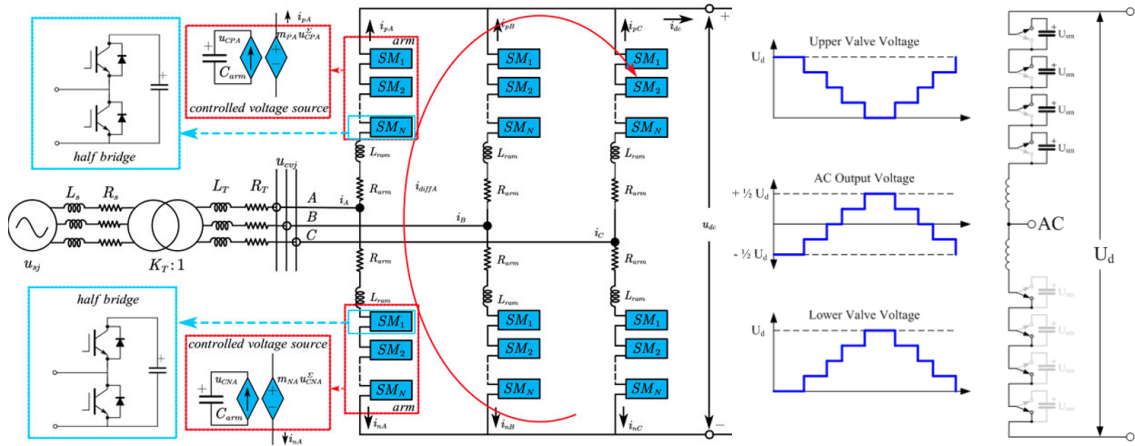


Figure 3: MMC Topology with Voltage Waveform

The MMC provides several capabilities that classical VSCs cannot. The small voltage steps greatly reduce harmonic content and dv/dt at the output. The modular structure allows the converter to scale to very high voltages without complex device stacking. The distributed energy storage in the sub-modules enables internal energy balancing and circulating current control. Finally, the topology increases availability, since sub-modules can be bypassed in case of equipment failure without shutting down the entire converter.

2.3 MMC Control Model

2.3.1 Why an Average Arm Model?

A detailed MMC model represents the switching state of every sub-module. While this gives a very accurate description of the converter, it quickly becomes impractical for realistic systems. The number of states grows with every added sub-module, the switching creates fast transitions that require very small simulation time steps, and the computational effort becomes too high for control development or system-level studies.

The Average Arm Model (AAM) offers a good alternative. Instead of modeling every sub-module individually, each arm is replaced by a controlled voltage source that represents the averaged behavior of the inserted sub-modules. This greatly reduces the size of the model and allows simulations to run much faster.

To ensure that the important internal dynamics are still captured, the AAM introduces a controlled current source, voltage source, and an equivalent arm capacitor. Together, these elements reproduce the combined charging and discharging behavior of all sub-module capacitors. This is shown in the red square in Figure 3. As a result, the model is able to represent both the external AC behavior and the internal circulating currents and energy variations that are essential for control design [10].

2.3.2 PR in the Stationary Frame

Proportional–Resonant (PR) controllers are used to improve the reference-tracking performance of the converter and to avoid several well-known limitations of conventional PI controllers. A classical PI controller cannot track sinusoidal signals in steady state without transformation to the rotating $d-q$ frame, and in single-phase systems it inherently produces a steady-state error. In contrast, the PR controller introduces a resonant term at the fundamental frequency, which allows it to achieve zero steady-state error directly in the stationary $\alpha\beta$ frame.

By operating in the stationary frame, the PR structure removes the need for a $d-q$ transformation and its filter-based decoupling terms. This makes the control scheme more flexible, especially under unbalanced or distorted grid conditions where the synchronous transformation is less effective [11].

2.3.3 Angle Estimation via SOGI

Since the control structure operates entirely in the stationary $\alpha\beta$ frame with PR current control, a conventional PLL is not necessary. Instead, a Second-Order Generalized Integrator (SOGI) is used in conjunction with a conventional PLL. The SOGI provides clean fundamental α - and β -components of the grid voltage and delivers a robust estimate of both the magnitude and the instantaneous phase. Because it naturally rejects harmonics and maintains good performance under unbalanced conditions, the SOGI-PLL offers a reliable way to obtain the angle and frequency information required for current reference calculations [12], [13].

2.3.4 Summary and Outlook

Together, the average arm model, PR current control, SOGI-based angle & magnitude estimation, and the required power-domain transformations form the conceptual foundation for the full control framework. The next chapter develops each of these elements in detail and derives the mathematical formulation for a complete MMC control system.

3 Theory

The foundation of the MMC is operation as a VSC at higher voltage conditions through the combined control of as many as hundreds of sub-modules [6]. This section discusses control based on an average model of these sub-modules, similar to VSC control with the added complication of sub-module energy management.

3.1 AAM

The AAM represents the voltage source behavior of an MMC arm while accounting for the energy storage property of the sub-module capacitance [14]. The model condenses the sub-modules of an arm into a dependent current source looped with equivalent capacitance behind a dependent voltage source representing the arm's output to the greater system.

The current through each sub-module capacitance is defined in terms of the total arm voltage u_{tot} , individual capacitance C_0 , and the number of sub-modules N as

$$C_0 \frac{du_{C_{tot}}}{dt} = \sum_{i=1}^N \frac{du_{C_i}}{dt} \quad (1)$$

Splitting the expression based on the number of activated sub-modules n yields

$$C_0 \frac{du_{C_{tot}}}{dt} = \sum_{i=1}^n \frac{du_{C_i}}{dt} + \sum_{i=n+1}^N \frac{du_{C_i}}{dt} \quad (2)$$

Given that the disconnected sub-modules do not conduct current and $C_{eq} = \frac{C_0}{N}$, Equation 2 can be simplified:

$$C_0 \frac{du_{C_{tot}}}{dt} = n \cdot i_{arm} \quad (3)$$

$$\frac{C_0}{N} \frac{du_{C_{tot}}}{dt} = \frac{n}{N} \cdot i_{arm} = i_{C_{eq}} \quad (4)$$

Finally, assuming equal capacitor voltages among the modules gives

$$u_{areq} = \frac{n}{N} \cdot u_{C_{tot}} \quad (5)$$

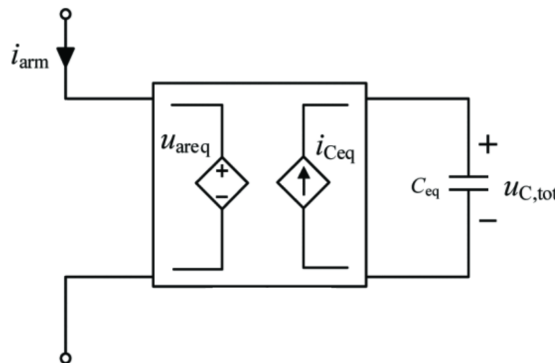


Figure 4: Average Arm Model Structure

Equations 4 and 5 thus provide the relationship between the dependent sources of the AAM shown in Figure 4. The dependent voltage source presents only the voltage of activated sub-modules to the external circuit, and the current from the external circuit only flows into activated sub-modules. The AAM allows accurate representation of both the energy stored in the arm capacitors through $u_{C_{tot}}$ as well as the voltage applied by the arm in terms of the modulation index $\frac{n}{N}$.

3.2 Proportional Resonant Control

The current control loop's function in the MMC is analogous to that of the VSC. They are both the innermost control loop that set the converter voltage so the current matches the input references. This can be done for three phases in either a stationary Clarke or rotating Park frame. This project employed a stationary reference frame to avoid decoupling the Park variables with filter parameters.

$$\begin{bmatrix} i_\alpha \\ i_\beta \\ i_0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & \sqrt{3} & -\sqrt{3} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (6)$$

The α and β Clarke components derived in Equation 6 are sinusoidal by nature of the stationary frame. In order to track a sinusoidal reference, the Proportional Resonant (PR) controller replaces the traditional PI controller by replacing the integral term with a second-order pole at the fundamental frequency $\omega = 50$ Hz. The infinite gain at 50 Hz allows the controller to follow the sinusoidal references.

$$K_p + \frac{2K_i}{s^2 + \omega^2} \quad (7)$$

However, the infinite gain at ω is not necessarily practical, as even a small disturbance in the system's operating frequency would greatly reduce the effectiveness of the controller. To rectify this, the gain peak is given a small bandwidth w_c to maintain a high gain at normal operating frequencies [11].

$$K_p + \frac{2K_i\omega_c s}{s^2 + 2w_c s + w^2} \quad (8)$$

Given an $w_c = 0.1$ Hz, the Bode magnitude plots of each controller are shown in Figure 5. The non-ideal controller with a bandwidth has a much lower gain at the resonant frequency but maintains a high gain for a bigger range.

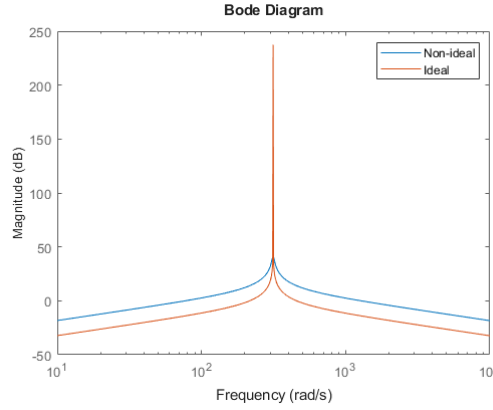


Figure 5: PR Controller Gain Magnitude

3.3 Power Transformation

The current references sent to the PR are calculated from the power references. The readily available power measurements are the power in each phase S_{abc} . Using the same Clarke transformation as Equation 6, the phase powers can be expressed in terms of Clarke current and voltage components.

$$\begin{bmatrix} S_a \\ S_b \\ S_c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{4} & -\frac{\sqrt{3}}{4} & -\frac{\sqrt{3}}{4} & \frac{3}{4} \\ \frac{1}{4} & \frac{\sqrt{3}}{4} & \frac{\sqrt{3}}{4} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} i_\alpha v_\alpha \\ i_\alpha v_\beta \\ i_\beta v_\alpha \\ i_\beta v_\beta \end{bmatrix} \quad (9)$$

Assuming v_α and v_β can be measured, current references could be calculated using Equation 9. However, the apparent power of each phase are not particularly useful quantities to control. Each current component has a phase and a magnitude, meaning there are four degrees of freedom for controlling the power with the current loop. The total active and reactive power constitute the ancillary services of the MMC, and two inter-arm active power quantities will allow the controller to manage the energy stored in each arm. To extract these from Equation 9, the following transformation is first applied:

$$\begin{bmatrix} S_{tot} \\ S_{a-b} \\ S_{a-c} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} S_a \\ S_b \\ S_c \end{bmatrix} \quad (10)$$

The result is the target apparent powers in terms of the Clarke components. They are separated into active and reactive power by splitting each component into its phase and magnitude as shown below:

$$\begin{aligned} S_{tot} &= \frac{3}{2}i_\alpha v_\alpha + \frac{3}{2}i_\beta v_\beta \\ P_{tot} &= \frac{3}{4}|v_\alpha||i_\alpha|\cos(\phi_{v_\alpha} - \phi_{i_\alpha}) + \frac{3}{4}|v_\beta||i_\beta|\cos(\phi_{v_\beta} - \phi_{i_\beta}) \\ Q_{tot} &= \frac{3}{4}|v_\alpha||i_\alpha|\sin(\phi_{v_\alpha} - \phi_{i_\alpha}) + \frac{3}{4}|v_\beta||i_\beta|\sin(\phi_{v_\beta} - \phi_{i_\beta}) \end{aligned}$$

Repeating this process for P_{a-b} and P_{a-c} results in a system of four equations with four unknown variables which can be solved to produce the current references given

power set points and voltage measurements. In order to solve the system more easily at simulation runtime, the coordinates are changed from polar to rectangular to remove the nonlinear functions. After expanding each term with the angle difference identities, the following substitutions are made:

$$\begin{aligned} |i_\alpha| \sin(\phi_{i_\alpha}) &= i_{\alpha y} \\ |i_\alpha| \cos(\phi_{i_\alpha}) &= i_{\alpha x} \\ &\dots \end{aligned}$$

with identical substitutions for i_β , v_α , and v_β . With the power equations in this form, an invertible matrix A in terms of the rectangular Clarke voltage components can be computed such that

$$[A] \begin{bmatrix} P_{tot} \\ Q_{tot} \\ P_{a-b} \\ P_{a-c} \end{bmatrix} = \begin{bmatrix} i_{\alpha x} \\ i_{\alpha y} \\ i_{\beta x} \\ i_{\beta y} \end{bmatrix} \quad (11)$$

which is implemented in a Simulink block to calculate the current references.

3.3.1 SOGI

The previous section requires the use of the PCC voltage measurements. In order to extract the magnitude and phase of the single-phase Clarke voltages, a Second-Order Generalized Integrator (SOGI) PLL is used.

In a dq0-controlled VSC, the PLL creates the phase angle of the Park transformations using a PI controller to lock the q-component to 0, since the q-component represents the phase of the transformed system. This technique can be applied to a single phase quantity by creating a quadrature signal to be used in the PLL's Park transform. There are several techniques to accomplish this, including adding a time delay directly or adding phase shifts through filters. The SOGI uses a notch filter for the grid input and a second order low-pass filter that shifts the phase by 90 degrees at the fundamental frequency, which is more complicated to implement but has better harmonic rejection [12]. The SOGI also includes a control parameter k which allows for tuning the dynamic response. The transfer functions and Bode responses of the SOGI from the single-phase quantity v_g to the direct and quadrature components v_α and v_β are shown below.

$$\frac{v_\alpha(s)}{v_g(s)} = \frac{k \omega s}{s^2 + k \omega s + \omega^2} \quad (12)$$

$$\frac{v_\beta(s)}{v_g(s)} = \frac{k \omega^2}{s^2 + k \omega s + \omega^2} \quad (13)$$

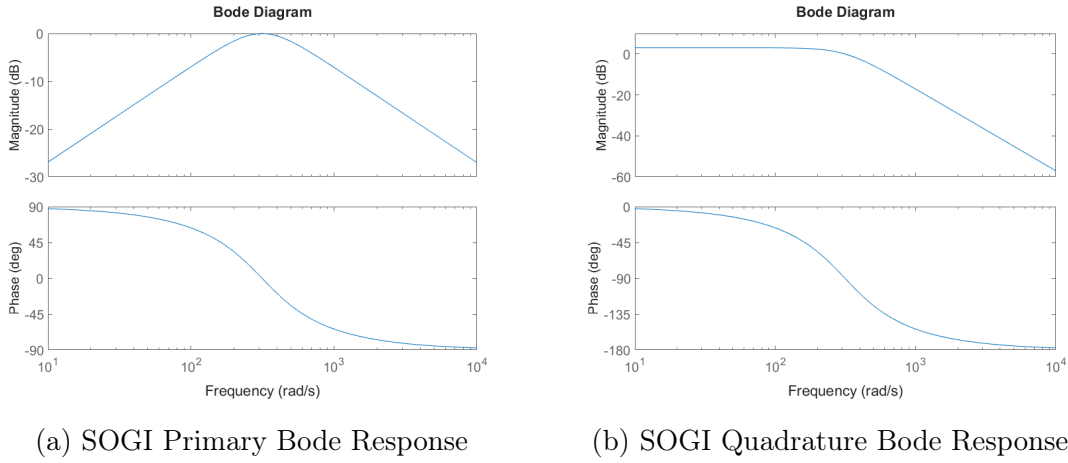


Figure 6: SOGI Bode Responses

The behaviors in Figures 6a and 6b make clear how the SOGI creates a filtered copy of the grid input signal and a 90° phase-shifted signal at the fundamental frequency, shown in the figures at 50 Hz.

These two outputs of the SOGI are then used in a typical PLL with a transform from Clarke $\alpha\beta 0$ to Park $dq0$. One SOGI-PLL each is used for v_α and v_β . The d component is the magnitude and the phase is tracked from the q component PI, which are used to calculate the current detailed in the previous section.

3.4 Energy Control

The outer control loop is used to maintain the balance of energy stored in each arm of the MMC. This is accomplished with a PI controller with energy references as an input and outputting to the power references. Since the powers that are controlled are the total power and the powers between Phase A and the other two phases, the energy references must also be the total and the difference in energy. Assuming equal energy in each arm, an equivalent capacitance of C and a DC voltage of V_{DC} , the energy references are

$$\begin{aligned} E_{tot}^* &= \frac{1}{2} C V_{DC}^2 \\ E_{a-b}^* &= 0 \\ E_{a-c}^* &= 0 \end{aligned}$$

These references will set the total power to exchange with the grid as necessary to maintain nominal energy and between arms to maintain parity with each other. This MMC model focuses on application as a STATCOM, so the fourth power reference Q_{tot} can be determined as necessary for the grid.

The outer control loop of a VSC must be designed to be slower than the inner control loops so as to prevent interference between their control actions. In this case, the PR controller inherently does not have a first-order response and therefore its time constant cannot be compared directly to that of the outer PI. The energy controller's time constant was tuned to be slower than the current controller based on the amplitude of the PR error's oscillations as an approximation for a first-order response.

4 Modeling Methodology

This chapter describes the modeling approach used throughout the project. All simulations were done in MATLAB/Simulink, gradually expanding the system as new parts are added. The modeling process was iterative and the model developed step by step as the understanding of the MMC and its control grew.

Many modeling choices were made in close consultation with the project supervisor. These decisions often concerned the level of model detail and bug-fixing. The objective was to create a model that is sufficiently detailed to capture the key MMC dynamics, while remaining practical to simulate and suitable for understanding and proving the control concepts as shown.

The remainder of this chapter presents the modeling steps, assumptions, and implementation details that form the basis for the results and analysis in the subsequent chapter.

4.1 Average Arm Model (AAM)

The initial model constructed in Simulink was the Average Arm Model (AAM) connected to an inductive load. This setup allowed the basic arm behavior and current flow to be tested before any higher-level control structures were added. The dependent sources inside the AAM directly implement the relations in Equations (4) and (5), which govern the averaged arm voltages and equivalent arm currents.

Figure 7 shows the complete AAM connected to a grid through an inductor. A modulation block generates the required arm voltage reference from the modulation index and the capacitor voltage. The resulting arm voltage is applied to the load, and the output voltage, inductor current, and grid voltage are measured for later use in the control loops and power calculation blocks.

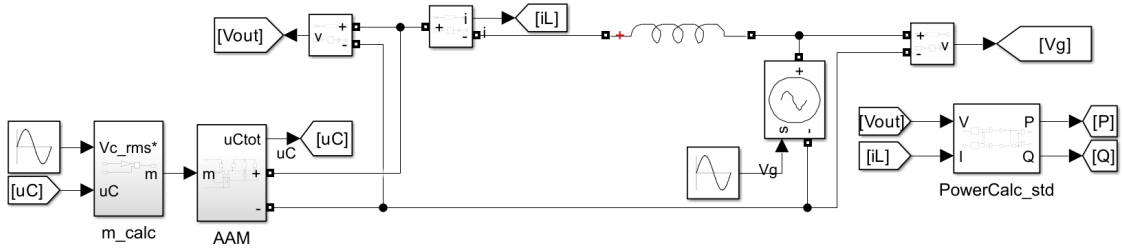


Figure 7: Single-Phase AAM Power Transfer Model

The internal structure of the AAM subsystem is shown in Figure 8. In the full AAM formulation, the equivalent arm capacitor would normally be included so that the model can reproduce the charging and discharging of all sub-module capacitors. However, at this stage of the project, no energy-balancing control was implemented yet. For this reason, the capacitor was temporarily replaced by a DC voltage source. This allowed the model to run without instability while still respecting the averaged arm-voltage behavior.

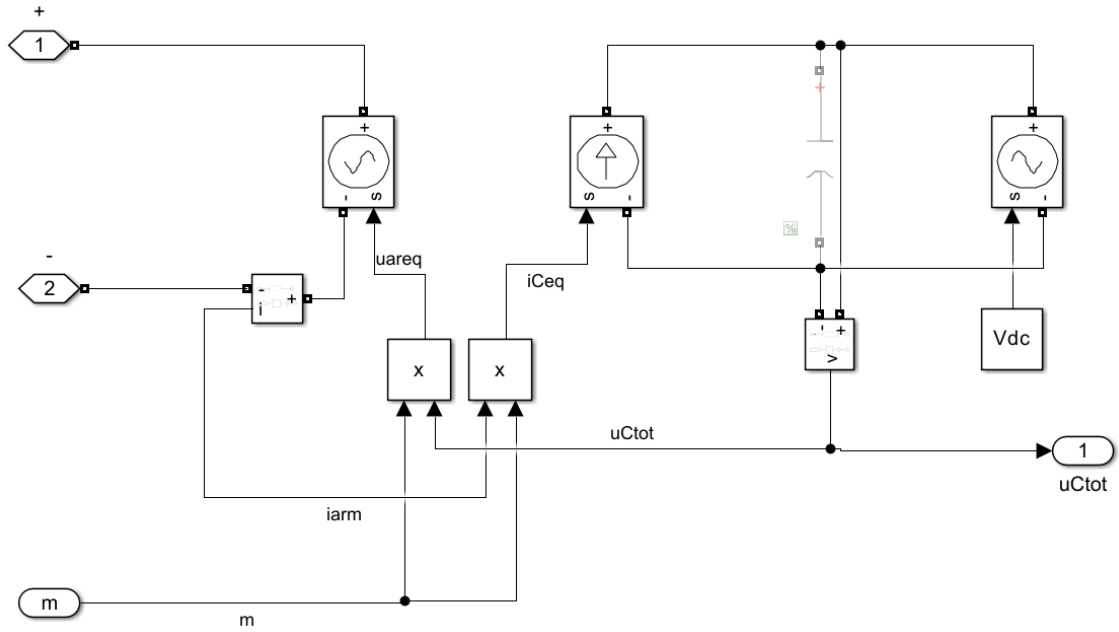


Figure 8: AAM Internal Subsystem

This simplified AAM setup provided a stable environment to test the modulation, current flow, and basic voltage synthesis of the MMC. Once these building blocks were confirmed to behave correctly, the model was gradually expanded with the additional control layers presented in the next sections.

4.2 Switching Model

To verify the behavior of the MMC arm with explicit device switching, a switching-based model was built. The overall structure of this model is identical to the AAM setup shown earlier in Figure 7. The only difference is that the AAM block is replaced by the switching arm model. This allows a direct comparison between averaged and switching behavior under the same external conditions.

Figure 9 shows the internal structure of the switching arm. Each arm consists of a full-bridge implemented using four IGBTs with anti-parallel diodes. The arm voltage is synthesized through PWM, based on the modulation index and a carrier waveform.

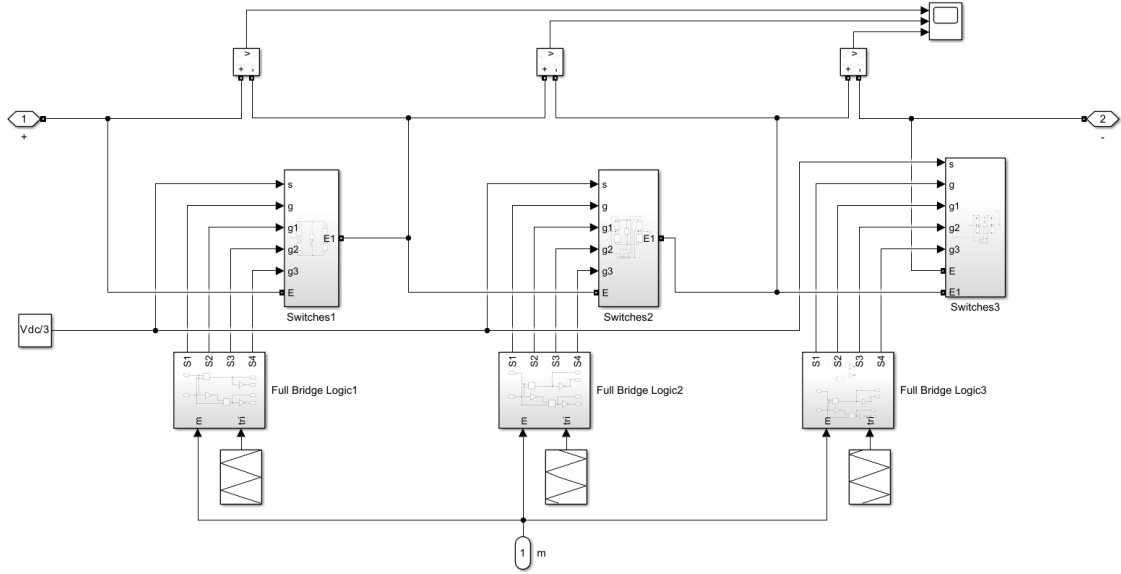


Figure 9: 3-Module Switching Model

The gate signals for the switches are generated by the full-bridge logic block, shown in Figure 10. The modulation index is compared with a triangular carrier to generate the four complementary PWM signals S_1 – S_4 . This forms the basic switching scheme.

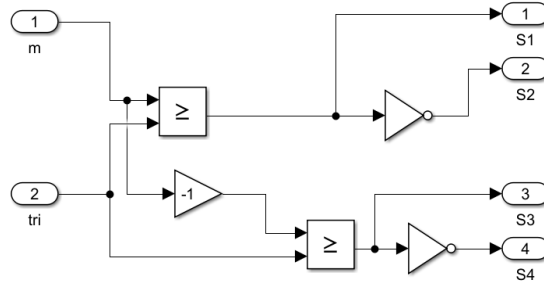


Figure 10: Full-bridge PWM Logic

The switching devices themselves are shown in Figure 11. Each leg consists of two IGBTs and their diodes, controlled directly by the logic outputs. Together, the four devices form the full-bridge that produces the instantaneous arm voltage.

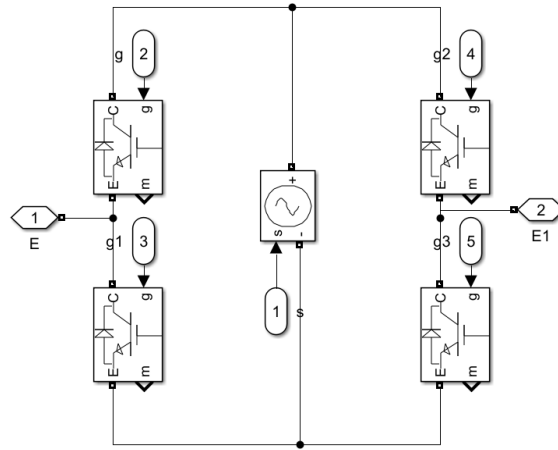


Figure 11: IGBT Configuration for Full-bridge

This switching model provides a more detailed representation of the converter, introducing switching ripple and dynamics that are not present while applying the AAM. Because the external structure remains unchanged, the averaged and switching models can be compared directly to understand the limitations and accuracy of the AAM.

4.3 STATCOM

After validating the individual converter models, the complete STATCOM model was assembled in Simulink. This model brings together the averaged converter, measurements, and all control blocks into one topology. Each of the three arms follows the AAM relations from Section 3.1: the arm voltage is determined through the dependent-source expression in (5), and the equivalent arm-capacitor current follows (4). The arms are connected to the grid through an RL filter, and all quantities are handled in the stationary $\alpha\beta$ frame using the Clarke transform in (6).

Figure 12 shows the full layout of the system, including the converter, the current controller, the SOGI blocks, and the power controller used to generate the reference currents.

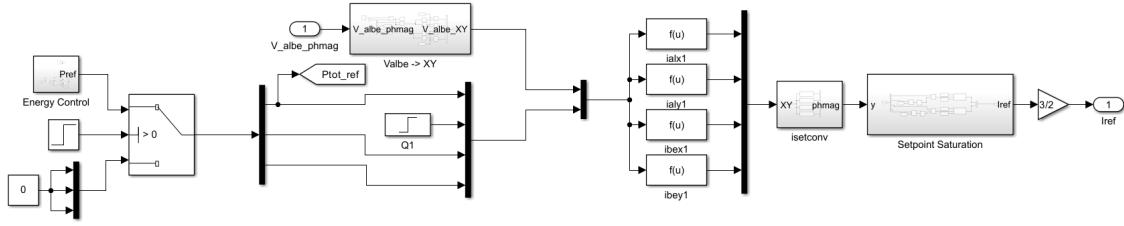


Figure 14: Power and Current Reference Calculation

Energy control. An outer energy-control loop stabilizes the stored energy in the converter arms. The references (E_{tot}^* , E_{a-b}^* , E_{a-c}^*) are determined as shown in subsection 3.4. This loop uses the relation between arm energy and capacitor voltage from (1)–(4) and provides an offset to P_{tot} , ensuring the arms remain balanced. The energy-loop bandwidth is kept slower than the current control, as required.

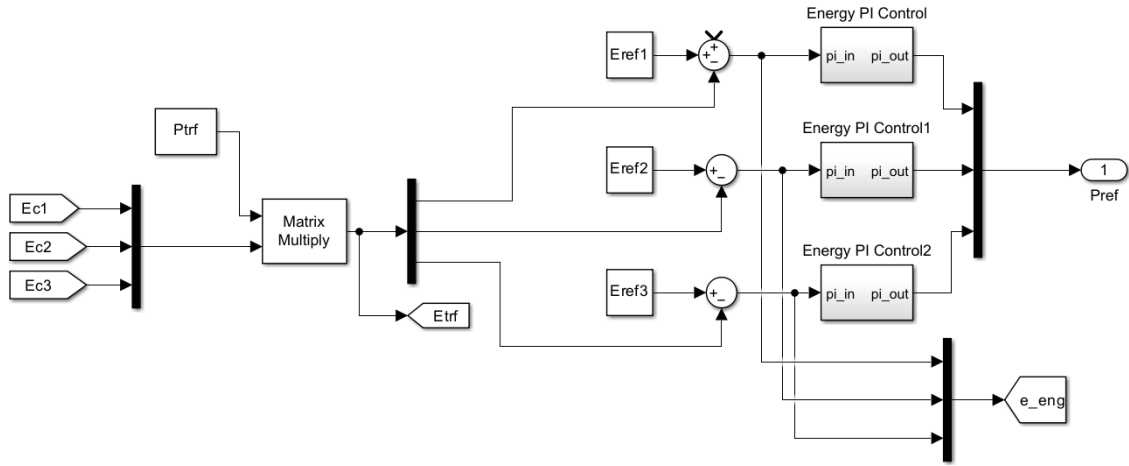


Figure 15: Energy Control Block

$\alpha\beta \rightarrow xy$ voltage transformation To support the power calculations, the measured Clarke voltages are rotated into an xy frame using the angle from the SOGI-PLL. This step uses the same $\alpha\beta$ quantities from (6) and provides the voltage components required for the rectangular power-current mapping.

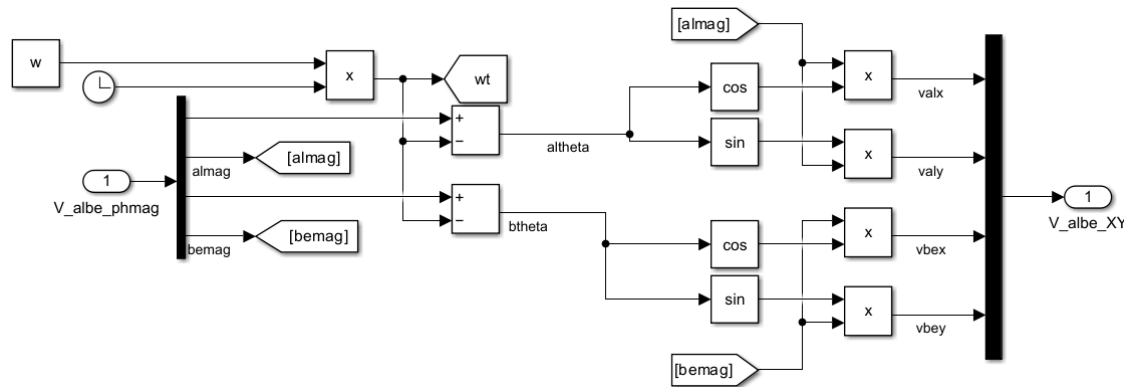


Figure 16: Transformation from Polar to Rectangular

4.3.3 PR Current Controller

The innermost loop is the PR current controller, which follows the non-ideal transfer function in (8). The controller tracks the reference currents i_α^* and i_β^* , and receives the measured currents through the Clarke transform. Its output forms the voltage reference sent back to the converter model.

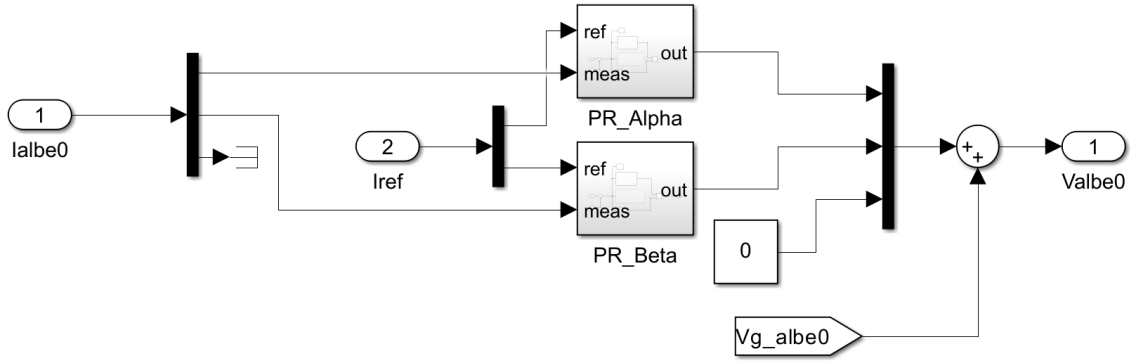


Figure 17: Current Controller

5 Results

This chapter presents the main simulation results. It is first validated that the Average Arm Model (AAM) and the switching model respond equivalently to identical power steps. Next, the validation experiments of the SOGI and PR control subsystems are presented. Finally, results for the complete STATCOM model are shown.

5.1 AAM vs. Switching Comparison

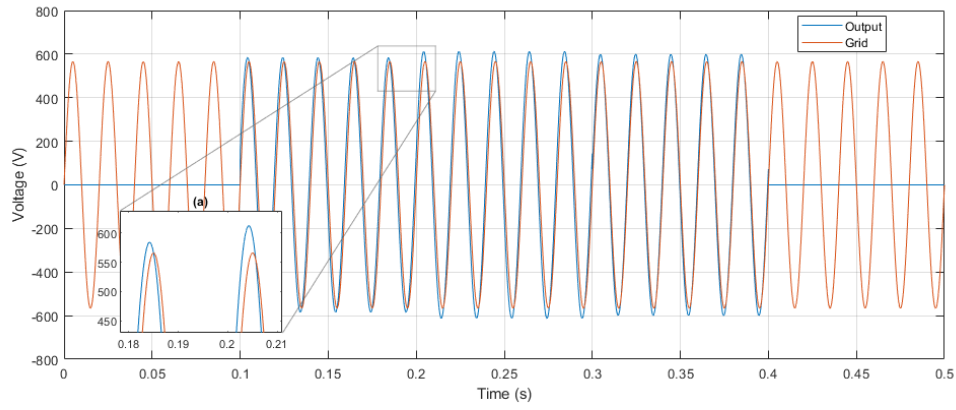
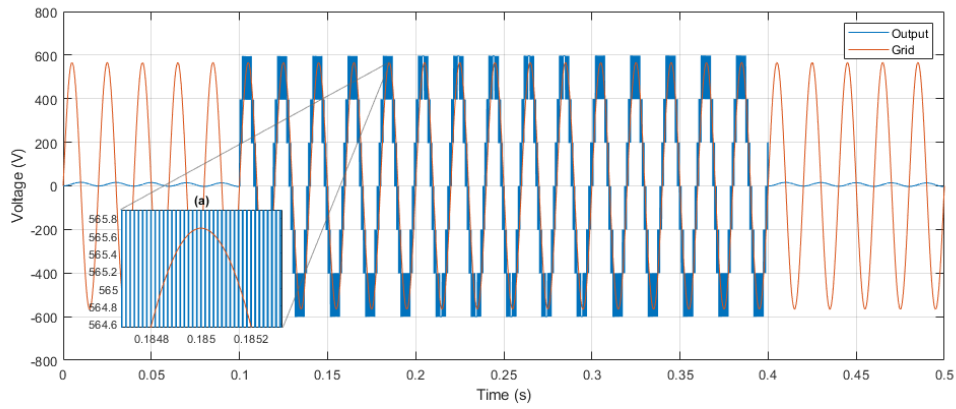
Both models are driven by the same step sequence of active and reactive power references in Table 1. The voltage magnitude and angle associated with these references is calculated using the method shown in A. The objective is to verify that the averaged representation and the explicit switching model exhibit the same macroscopic behavior. As shown below, they do: P tracks identically (within PWM ripple for the switching case), and a consistent offset appears in Q for both models. Because the Q error is identical in AAM and Switching, the issue likely originates in the shared power-calculation path or another block outside the converter substitution. In a closed-loop implementation (with the PR current and power controllers), this offset would be corrected; here we apply direct references and do not yet close the loop.

Time (s)	P (W)	Q (VAr)
0.0	0	0
0.1	100 000	0
0.2	100 000	20 000
0.3	50 000	20 000
0.4	0	0

Table 1: P&Q Reference Step Changes

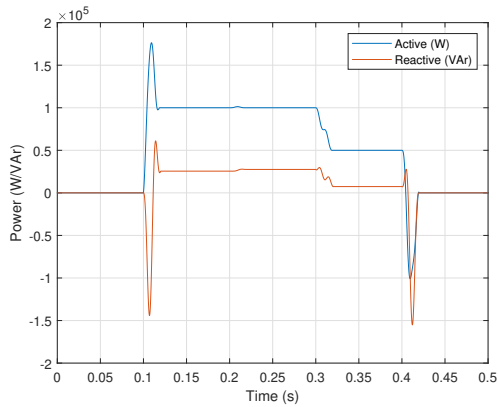
Voltages The AAM clearly shows an increase in output voltage magnitude after the Q step at $t = 0.2$ s, consistent with reactive power injection. The switching model shows the same trend, and additionally reveals the expected PWM staircasing: with three full-bridge submodules per arm, the arm/output voltage is synthesised in six small increments.

P & Q Active power follows the reference sequence in both models with comparable transients; the switching case shows the expected ripple due to PWM. The same steady offset appears in reactive power in both models, pointing to a shared cause outside the AAM/switching substitution. This will be addressed once the PR current controller and power controller are closed around the converter.

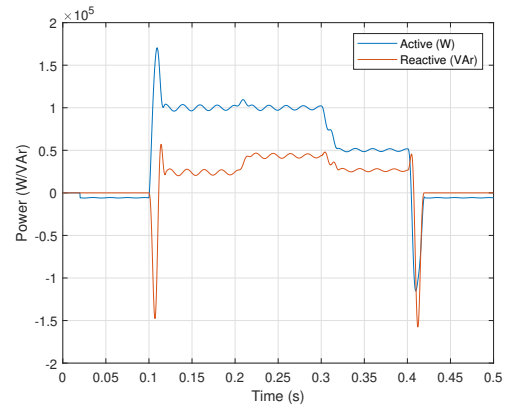
(a) AAM Voltage (zoom around $t=0.2$ s).

(b) Switching voltage (PWM steps visible).

Figure 18: Output vs. Grid Voltage for AAM and Switching.



(a) AAM active/reactive power.



(b) Switching active/reactive power.

Figure 19: Power Response under Table 1 Sequence

Overall, the results show that the AAM and switching models produce the same macroscopic active and reactive power behaviour under the applied steps. The switching model shows the expected six-level staircasing in the output voltage, cor-

responding to the three full-bridge submodules and the PWM scheme. A offset appears in the reactive-power response of both models, indicating that the cause lies in the shared measurement or transformation path, or in the absence of closed-loop control, rather than in the converter modelling itself. This deviation is expected to disappear once the PR current controller and power controller are included in the full STATCOM model.

5.2 PR

The PR controller was tested in isolation, since it is the lowest-level control loop of the converter and only works in conjunction with the inverse Clarke transformation. The error behavior of both an ideal and non-ideal PR controller were tested with simple sinusoids at 50 Hz and again with a 1% frequency error. The results are shown in Figures 20 and 21.

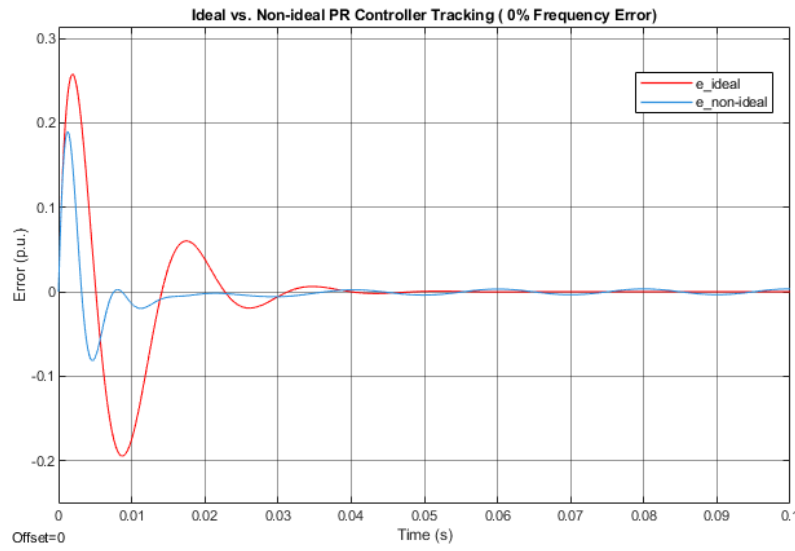


Figure 20: PR Response at Exact Frequency

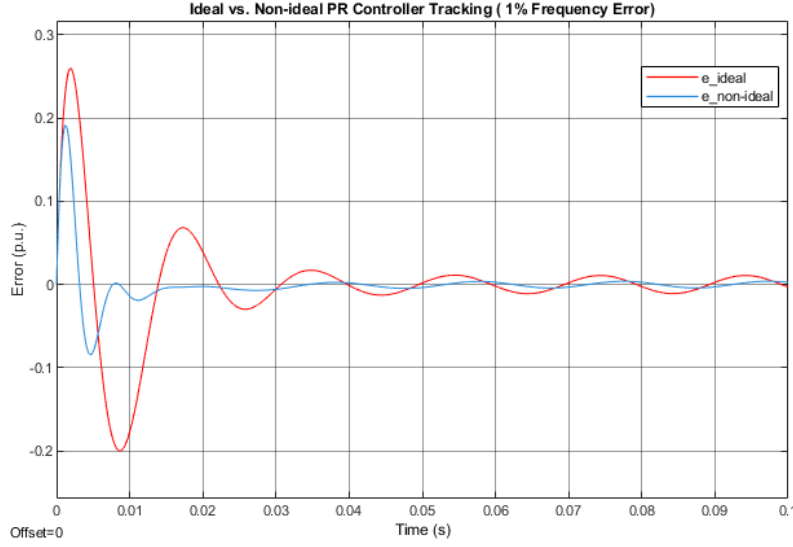


Figure 21: PR Response at Frequency Deviation

The K_i is equal to 120, and the bandwidth of the non-ideal controller is chosen to be much smaller than the frequency deviation to highlight the differences in performance. The plots show the non-ideal controller reducing to a smaller steady-state error in the deviated frequency condition. When using a more realistic bandwidth equal to normal grid frequency operating range in the three-phase model, both PR controllers performed similarly. The current response for the reactive power regulation in Section ?? is shown below.

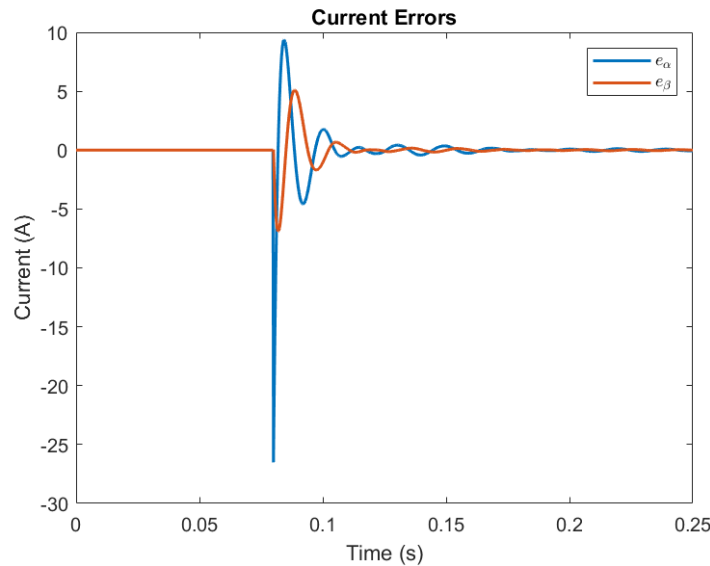


Figure 22: PR Current Controller Response

5.3 SOGI

The SOGI was implemented and tested in the three-phase model before the power calculation to determine the dynamics of its response, with dummy constants fed

to the current controller. Figure 23 shows the results of the two Clarke voltage components being tracked by the SOGI-PLL.

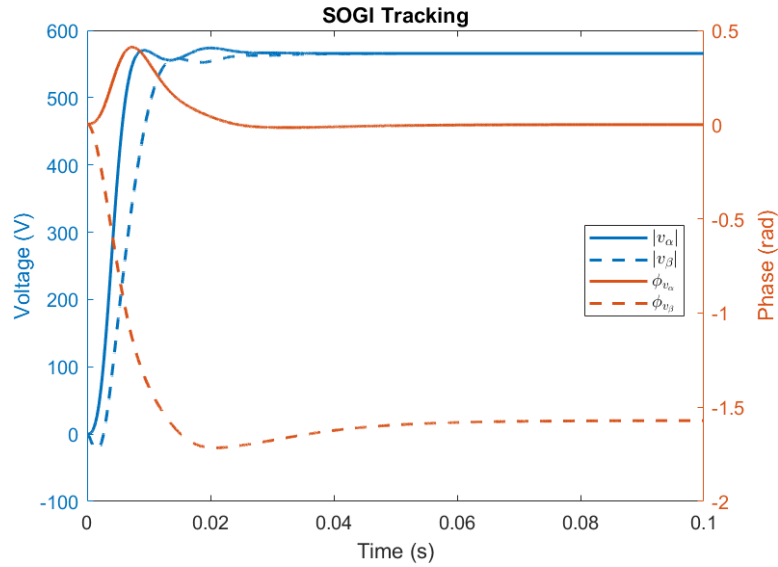


Figure 23: SOGI Tracking Response

The results show the two grid voltage components with identical magnitudes and a $\frac{\pi}{2}$ radian phase shift, confirming it is tracking correctly. The dynamics also settle out by 60 ms, after which is the time all the other converter controller dynamics should be tested to prevent interference.

5.4 STATCOM

Finally, the last experiments with the MMC model were on its application as a STATCOM device. The goal was to assess its capability to provide reactive power while maintaining the energy parity of the capacitors. Figures 24 and 25 show a step change of -15 kVAR in the reactive power reference applied at 80 ms.

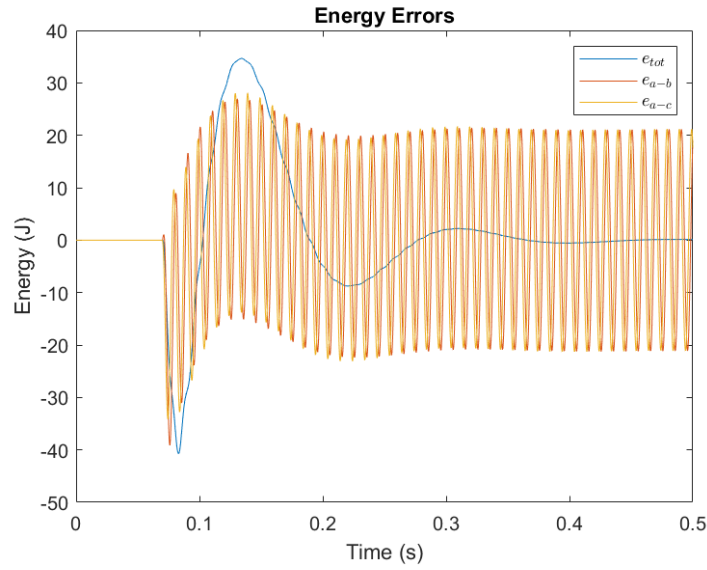


Figure 24: Capacitor Energy Balance

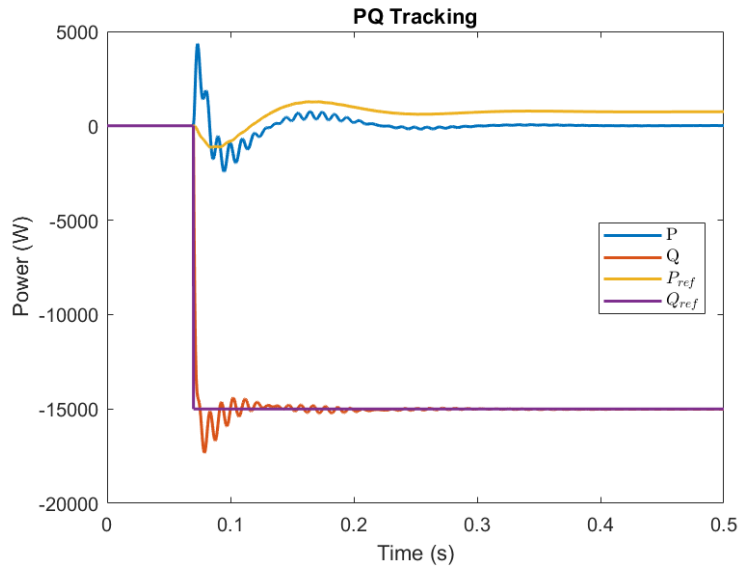


Figure 25: PQ Tracking

The power results show that the reactive power tracks properly with oscillations at the grid frequency caused by the resonating inherent to the current controller. The active power shows a constant error with a non-zero reference. The discrepancy is analogous to those observed in both the AAM and switching model power tracking cases, which suggests some more fundamental coupling issue between the active and reactive power stemming from the model. Given the settling of the total energy error to zero in Figure 24, however, it seems that the actual power output of the converter is correct.

The energy results show that the total energy of the capacitors settles to the nominal value as well as a steady-state oscillation in the energy between the branches.

[15] suggests that the capacitor energy oscillations at the grid frequency are expected and the controller can incorporate a notch filter to avoid compensating these oscillations.

6 Discussion and Conclusion

This report discussed the procedure of building a Modular Multilevel Converter model in MATLAB/Simulink based on the Average Arm Model architecture. First, the background of each necessary component to be developed is introduced, then described mathematically, and finally the results of its implementation are presented. The AAM composes the base approximation for an MMC due to its inherent capacitance, a PR current controller is used to set the AAMs' modulation indices in the Clarke frame, a SOGI-PLL extracts the necessary information to generate four current references from four controlled system powers which are set by an outer PI controller to regulate the energy of the system.

The model had several areas for improvement that could be pursued. There were numerous challenges testing the control loops with step disturbances, since introducing discontinuities into the Simulink algebraic loops made it quite sensitive to failure. Features like feed-forward terms and unbalanced grid conditions became much more challenging to experiment with due to these difficulties. Further development of the model could include functionality beyond a STATCOM by including a representation of a DC system, which would allow the MMC to act as a more common VSC. The control could also be separated into AC and DC components and into upper and lower branch arms in order to control unbalanced conditions.

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A Appendix A: Voltage Magnitude and Angle Calculation

Consider the standard voltage-behind-reactance model, where the converter internal voltage magnitude is V_1 , the grid voltage magnitude is V_2 , and the line reactance is $X > 0$. The active and reactive power delivered to the grid are

$$P^* = \frac{V_1 V_2}{X} \sin \delta, \quad (14)$$

$$Q^* = \frac{V_1 V_2}{X} \cos \delta - \frac{V_2^2}{X}, \quad (15)$$

where $\delta = \theta_1 - \theta_2$ is the power angle.

Define the auxiliary variables

$$A := XP^*, \quad B := XQ^* + V_2^2. \quad (16)$$

Then the power expressions become

$$A = V_1 V_2 \sin \delta, \quad (17)$$

$$B = V_1 V_2 \cos \delta. \quad (18)$$

Solution for the Required Voltage Magnitude

Squaring and adding the above equations yields

$$A^2 + B^2 = (V_1 V_2)^2. \quad (19)$$

Thus, the required converter internal voltage magnitude is

$$V_1 = \frac{\sqrt{A^2 + B^2}}{V_2} = \frac{1}{V_2} \sqrt{(XP^*)^2 + (XQ^* + V_2^2)^2}. \quad (20)$$

Solution for the Power Angle

The power angle follows directly from

$$\delta = \text{atan2}(A, B) = \text{atan2}(XP^*, XQ^* + V_2^2). \quad (21)$$

In typical AAM/GFM operation, the principal-value solution with the smallest $|\delta|$ is selected.